Data-Driven Sector ETF Rotation Strategy

— A Quantitative Research Framework Based on Large-Scale Signal Mining and Machine Learning

> **Core Positioning**: This project replaces subjective judgment and single hand-picked indicators with a systematic, data-driven pipeline. We screen 80+ price-volume factors and multiple ML models under a unified quantitative standard, automatically identifying effective signals, then deploy them through a trading execution framework to capture Alpha. The entire research process is data-driven, reusable, iterable, and extensible.

Audience: Finance Industry Professionals
Date: 2026-05-13
Backtest Period: 2025-01-01 to 2026-01-31 (57 weekly periods, zero fees, zero slippage)

I. Research Motivation

1.1 Market Background

Sector rotation in US equities is well-established. Different phases of the economic cycle correspond to different sector leadership — defensive sectors lead in recessions, cyclical sectors surge in recoveries. **SPDR Sector ETFs** are the most direct instrument for capturing this pattern: 11 ETFs cover all GICS Level-1 sectors (Technology, Financials, Energy, Healthcare, Consumer, etc.), with strong liquidity and low trading costs — naturally suited for rotation strategies.

1.2 Limitations of Traditional Approaches

Most sector rotation strategies rely on **a handful of manually constructed momentum indicators**, such as 3-month or 6-month price momentum. These approaches have two fundamental flaws:

- **Subjective factor selection**: researchers pick indicators based on intuition, with no systematic way to assess which factors genuinely have predictive power
- **No framework for parameter tuning**: the combination space of lookback windows, position counts, and other parameters is enormous — manual search is both inefficient and prone to overfitting

1.3 Our Starting Point

Our goal is to build a **systematic, iterable quantitative research framework**:

> Use a unified evaluation standard to objectively test a large number of signals, and let the data tell us which signals are effective — rather than guessing.

This framework not only answers "which factor is best," but also lays the foundation for continuous iteration (adding new factors, optimising combination weights, incorporating machine learning).

II. Research Framework Overview

Core Question: What kind of signal can accurately predict the relative strength of sector ETFs over the next 4 weeks, after each Wednesday close?

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┌─────────────────────────────────────────────────────────────────────┐
│                    Raw Data (Weekly Update)                         │
│          11 Sector ETFs  ×  OHLCV + Total Return Index (TRI)        │
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                        │                          │
          ┌─────────────▼─────────────┐  ┌─────────▼─────────────────────┐
          │    Alpha Factor Factory   │  │        ML Prediction Pipeline  │
          │                           │  │                                │
          │  101 Formulaic Alphas     │  │  Labels: FRS (4-week returns)  │
          │  + 29 Andy Factors        │  │  ┌──────────────────────────┐  │
          │         ↓                 │  │  │ FRS1  Total Return       │  │
          │  IC Pre-Screen            │  │  │ FRS2  Sharpe Proxy       │  │
          │  (Mean IC > 0.02)         │  │  │ FRS3  Vol-Penalised Ret. │  │
          │         ↓                 │  │  └──────────────────────────┘  │
          │  82 Viable Alpha Factors  │  │         ↓ Train on FRS labels  │
          └─────────────┬─────────────┘  │  LightGBM / XGBoost / MLP /    │
                        │                │  PCA+Ridge / Ensemble          │
                        │                │         ↓                      │
                        │                │  5 ML Signals (weekly ETF scores) │
                        │                └────────────────┬───────────────┘
                        │                                 │
                        └──────────────┬──────────────────┘
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                     ┌─────────────────▼─┐
                     │  Signal Pool      │
                     │ 82 Alpha + 5 ML   │
                     └─────────────────┬─┘
                                       │
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         ┌─────────────────────────────▼──────────────────────────────┐
         │                  Layer 1: Signal Evaluation Framework        │
         │                                                             │
         │  Baseline: Equal-Weight (Sharpe 1.305) — mandatory floor    │
         │                                                             │
         │  Each signal → Softmax weights → Long-only backtest → Sharpe│
         │                                                             │
         │  Screening Results:                                         │
         │    Best Alpha:  Alpha#19  (Sharpe 1.968)                    │
         │    Best ML:     Signal 2 Ensemble  (Sharpe 1.506)           │
         └─────────────────────────────┬──────────────────────────────┘
                                       │ Top signals passed down
         ┌─────────────────────────────▼──────────────────────────────┐
         │                  Layer 2: Signal Deployment                  │
         │                                                             │
         │  Market-Neutral Long-Short Strategy                         │
         │  Long Top-3  ←── Signal Ranking ──►  Short Bottom-3         │
         │                                       (momentum filter)     │
         │  Risk Control: 3-Tier State Machine (NORMAL/LIGHT/HEAVY)    │
         │                                                             │
         │  Deployment Results:                                        │
```

```
│    Best Alpha:  Alpha#23  (Sharpe 2.081, Beta 0.234)          │
│    Best ML:     Signal 2  (Sharpe 0.806)                      │
└───────────────────────────────────────────────────────────────┘
```

Research Logic: Layer 1 standardises the comparison environment to screen signals at scale (signal predictive power in isolation). Layer 2 validates how signals perform inside a real trading framework. The two layers are decoupled — a signal must prove itself in both dimensions to be considered production-ready.

III. Signal Pool

Before explaining how we evaluate signals, here is what we are testing.

3.1 Price-Volume Alpha Factors

Source: 101 Formulaic Alphas

The core of the factor library comes from the academic paper *101 Formulaic Alphas* (Kakushadze, 2015), a widely cited factor set in Wall Street quant practice. Of the original 101 factors, **82 implementable factors** are retained — the remaining 19 require market-cap data or industry neutralisation (meaningless in an 11-ETF universe).

Economic Classification

Category	Representative Factors	Core Logic
Momentum / Trend	Alpha#19, #24, #37	Trend continuation — strong sectors stay strong
Mean Reversion	Alpha#23, #9, #49	Price recovery after short-term over-extension
Volume-Price	Alpha#3, #12, #13	Volume-price divergence or co-movement signals
Vol-Adjusted	Alpha#34, #35, #55	Strip out high-volatility noise, isolate trend

Andy Factors

Beyond the 101 formulaic factors, we developed **29 custom factors (Andy Factors)** designed specifically for sector ETF rotation. The most critical:

- **12-Week Cumulative Return Factor**: measures absolute momentum direction. Used in the baseline strategy as the **short-side filter** — only ETFs with negative absolute momentum (downward trend) are eligible for shorting, preventing short positions against rising sectors.

IC Pre-Screening

All factors pass through an **Information Coefficient (IC)** filter before entering the full backtest. IC measures rank correlation between predicted and realised returns: higher mean IC means more accurate directional predictions. Only factors with Mean IC > 0.02 and IC-IR > 0.3 enter the full backtest.

Top factors after pre-screening:

Factor	Mean IC	Economic Intuition
Alpha#50	0.042	Volume-VWAP correlation structure
Alpha#3	0.034	Open-volume negative correlation
Alpha#41	0.034	Geometric mid-price vs VWAP spread
Alpha#24	0.033	100-day moving average momentum
Alpha#98	0.031	VWAP × short-term turnover interaction

3.2 Machine Learning Signals

Why Machine Learning

Traditional factor combinations depend on researcher judgement, and parameter search requires extensive manual trial-and-error. The core value of ML is:

> **Let the model automatically learn effective combination weights across all 82 factors, using forward returns as supervision labels — discovering non-linear relationships that humans cannot easily detect.**

Prediction Target Design (FRS)

ML models predict a **Future Return Score (FRS)** — each ETF's performance over the next 4 weeks. Three label variants target different investment preferences:

Label	Definition	Use Case
FRS1	4-week total return	Directional return prediction
FRS2	4-week Sharpe proxy	Risk-adjusted performance
FRS3	FRS1 − 2 × volatility	Penalises high-vol outperformance

Each model scores all 11 ETFs weekly — higher score means "more worth buying."

Five ML Signals

Signal	Model	Target	Approach
Signal 1	LightGBM	FRS3	Industry-standard gradient boosting for tabular data
Signal 2	Ensemble (rank-avg)	FRS1	Combines multiple models — reduces single-model bias
Signal 3	XGBoost	FRS3	Paired with LightGBM to validate model robustness
Signal 4	PCA + Ridge	FRS3	Linear baseline after dimensionality reduction
Signal 5	MLP	FRS2	Captures non-linear factor interactions

IV. Signal Evaluation Methodology

4.1 Why a Unified Evaluation Framework

The most direct way to evaluate a signal is to plug it into a real strategy and observe the resulting Sharpe. But this creates a problem: **if different signals use different execution logic, we cannot attribute performance differences to the signal itself vs. the strategy design.**

Our solution: design a **standardised evaluation strategy**, where all signals compete under identical execution logic — isolating each signal's pure information content.

4.2 Evaluation Strategy: Softmax Weight Allocation

Core Mechanism:

1. Signal scores all 11 ETFs (higher score = stronger expected performance)
2. Softmax converts scores to weights — higher-scoring ETFs get larger positions; all ETFs always held (fully invested long)
3. Weights rebalanced each week based on new signal

$$w_{\text{ETF}_i} = \frac{e^{s_i}}{\sum_{j=1}^{11} e^{s_j}}$$

Advantages:

- Weights change continuously — no hard "buy top N" cutoff that triggers excessive rebalancing at rank boundaries
- Signal strength maps directly to weight — comparison between signals is completely fair
- Naturally extensible to multi-signal weighted combinations

Weekly Allocation Flow

```
Every Wednesday after close
      │
      ▼
┌──────────────────────────────────────────────────────┐
│             Signal Scoring (11 ETFs)                 │
│                                                      │
│   XLK  ████████████████  +2.1  (Tech, Strong)        │
│   XLV  ████████████      +1.4  (Healthcare)          │
│   XLF  ████████          +0.9  (Financials)          │
│   XLI  ██████            +0.5  (Industrials)         │
│   XLB  ████              +0.2  (Materials)           │
│   XLP  ███               +0.0  (Staples)             │
│   XLY  ██                -0.3  (Discretionary)       │
│   XLC  ██                -0.5  (Communication)       │
│   XLRE █                 -0.8  (Real Estate)         │
│   XLU  █                 -1.1  (Utilities)           │
│   XLE                    -1.6  (Energy, Weak)        │
└──────────────────────────────────────────────────────┘
      │
      │  Softmax: high score → large weight, low score → small weight
      │  All weights sum to 100%, no market timing needed
      ▼
```

```
  ┌──────────────────────────────────────────────────────┐
  │                 Portfolio Weights                      │
  │                                                      │
  │   XLK  ████████████████   18.2%                      │
  │   XLV  █████████████      14.5%                      │
  │   XLF  ██████████         11.0%                      │
  │   XLI  ████████            8.6%                      │
  │   XLB  ███████             7.4%                      │
  │   XLP  ██████              6.8%  ← lower score =     │
  │   XLY  ██████              6.4%    smaller weight,   │
  │   XLC  █████               6.0%    but still held    │
  │   XLRE █████               5.5%                      │
  │   XLU  ████                4.8%                      │
  │   XLE  ████                4.4%                      │
  │        ─────────────────────────                     │
  │        Total               100% (fully invested)     │
  └──────────────────────────────────────────────────────┘
        │
        │  Execute rebalancing on Thursday close
        ▼
  Hold until next Wednesday close, repeat
```

> **vs. traditional top-N strategies**: A hard cutoff treats rank 3 and rank 4 identically and forces a full rebalance whenever they swap. In the Softmax strategy, a small rank movement causes only a small weight shift — naturally reducing turnover at rank boundaries.

4.3 Equal-Weight Baseline: The Mandatory Floor

Before evaluating any signal, we establish the **equal-weight portfolio** as the benchmark floor: hold all 11 ETFs at 1/11 each, every week, with no signal. This represents market average performance under zero information.

Any signal that cannot beat the equal-weight baseline is considered statistically uninformative.

Metric	Equal-Weight Baseline
Sharpe	**1.305**
Annual Return	13.98%
Annual Volatility	10.46%
Max Drawdown	−10.64%
Excess Return vs SPY	−3.08%

4.4 Three-Step Optimisation Roadmap

The sole evaluation metric is **Sharpe Ratio**.

Step	Content	Method	Status
Step 0	Equal-weight baseline (mandatory floor)	—	✅ Complete
Step 1	Single-signal screening: test each factor or ML signal individually	IC pre-screen + Sharpe ranking	✅ Complete
Step 2	Multi-signal linear blend: find optimal weighting	Bayesian optimisation + walk-forward CV	🔄 In Progress
Step 3	Neural network end-to-end learning	Gradient descent, Sharpe as training loss	⏳ Planned

4.5 Alpha Factor Screening Results (Step 1)

40 factors backtested under the unified evaluation framework:

Factor	Sharpe	Ann. Return	Ann. Vol	Max DD	vs Equal-Wt
Alpha#19 ★	**1.968**	**23.33%**	10.97%	**−7.26%**	**+0.663**
Alpha#24	1.758	24.60%	13.01%	−11.32%	+0.453
Alpha#23	1.665	18.78%	10.69%	−10.66%	+0.360
Alpha#31	1.561	16.57%	10.16%	−9.30%	+0.256
Equal-Weight (floor)	1.305	13.98%	10.46%	−10.64%	—
Alpha#101	1.108	11.73%	10.51%	−10.18%	−0.197

Best single factor: Alpha#19 (Sharpe 1.968, CAPM Alpha 11.35% vs SPY, Beta 0.613)

Alpha#19 is a **long-horizon momentum reversal signal** (250-day cumulative return direction + cross-sectional ranking) — takes contrarian positions when a sector shows extended outperformance.

> **Screening conclusion**: Of 40 factors tested, only a handful meaningfully beat the equal-weight floor. The majority are statistically indistinguishable from holding all 11 ETFs equally — confirming the necessity of systematic screening. Manual factor selection would almost certainly misjudge these results.

4.6 ML Signal Screening Results (Step 1)

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	vs Equal-Wt
Signal 2 (Ensemble) ★	**1.506**	**16.76%**	10.68%	−10.27%	**+0.201**
Signal 5 (MLP)	1.329	14.06%	10.31%	−10.26%	+0.024
Equal-Weight (floor)	1.305	13.98%	10.46%	−10.64%	—
Signal 3 (XGBoost)	1.304	13.97%	10.46%	−10.62%	−0.001
Signal 4 (PCA+Ridge)	1.304	13.96%	10.44%	−10.61%	−0.001

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	vs Equal-Wt
Signal 1 (LightGBM)	1.300	13.92%	10.46%	−10.64%	−0.005

Best ML signal: Signal 2 (Ensemble / FRS1), Sharpe 1.506, CAPM Alpha 4.52% vs SPY.

Key finding: The three FRS3-targeting models are statistically indistinguishable from equal-weight. Predicting vol-penalised returns is far harder than directional returns (FRS1). The ensemble's rank-averaging across models suppresses single-model noise — this is the path forward for ML signals.

> **Going forward**: Ensemble approach + directional label (FRS1) should be the primary ML signal direction.

V. Signal Deployment: Market-Neutral Long-Short Baseline Strategy

5.1 Strategy Positioning

The baseline strategy is our **actual trading execution layer**, designed as a **market-neutral long-short hedge**:

- Long the top-ranked (strongest) sector ETFs by signal
- Short the bottom-ranked (weakest) sector ETFs by signal
- Long and short positions are equal and offsetting — near-zero net exposure

This design strips out the effect of overall market direction (Beta) — P&L comes entirely from cross-sectional sector calls (Alpha).

5.2 Position Structure

Direction	Count	Weight per Position	Total
Long	3	+33.3%	+100%
Short	3	−33.3%	−100%
Gross Exposure			**200%**
Net Exposure			**~0%**

5.3 Short Filter Mechanism

Shorting is not unconditional. For bottom-ranked ETFs, one additional condition must be met:

> **12-week cumulative return must be negative** (absolute momentum pointing down)

If a bottom-ranked ETF is still up over the past 12 weeks, the short slot is left in cash. This prevents **shorting rising sectors** — even if a sector is relatively weak versus peers, forcibly shorting an absolute uptrend carries extreme risk.

5.4 Rank Stickiness

To avoid excessive rebalancing from minor rank fluctuations (e.g., rank 3 vs. rank 4 swapping):

- Long holdings: retained unless rank falls outside top **5** (±2 tolerance zone)
- Short holdings: closed only if rank rises above bottom **5**
- Exception: short positions with **absolute momentum turning positive are closed immediately**, regardless of stickiness

5.5 3-Tier Drawdown Risk Machine

The strategy has a built-in dynamic risk control mechanism that automatically adjusts gross exposure based on portfolio drawdown:

```
          DD ≥ 10%                  DD ≥ 15%
NORMAL ───────────► LIGHT ────────────────► HEAVY
(200% gross)       (100% gross)            (0%, full cash)
    ▲                   ▲
    │  DD < 8%          │  ≥ 2 proposed longs w/ positive
    │  for ≥ 2 weeks    │  absolute momentum, for ≥ 2 weeks
    └───────────────────┘
```

Recovery conditions are deliberately strict, requiring **consecutive weeks** of meeting the threshold — preventing premature re-entry after a brief bounce that leads to a second drawdown.

5.6 Baseline Strategy Backtest Results

Alpha Signals (40 factors fully tested)

Factor	Sharpe	Ann. Return	Ann. Vol	Max DD	Beta vs SPY	CAPM Alpha
Alpha#23 ★	**2.081**	**23.00%**	10.21%	**−5.85%**	**0.234**	**17.33%**
Alpha#57	1.528	17.72%	11.08%	−7.40%	—	—
Alpha#37	1.478	16.98%	11.03%	−5.78%	—	—
Alpha#10	1.230	13.37%	10.67%	−6.90%	—	—
Alpha#19 *(best in Softmax)*	−0.591	−5.66%	9.15%	−11.16%	—	—

Best result: L/S + Alpha#23 (Sharpe 2.081, CAPM Alpha **17.33%** vs SPY, Beta **0.234**)

Alpha#23 is a **near-peak breakout reversal signal**: when a sector's price breaks above recent highs, it anticipates short-term mean reversion and positions contrarily. Market Beta is only 0.234 — strategy returns are highly independent of overall market direction.

Note: **Alpha#19 (best in Softmax) produces negative Sharpe in the L/S framework**. This demonstrates that different signals have clear framework compatibility — the Softmax long-only framework tests absolute ranking ability, while the L/S framework demands two-way predictive accuracy (both strong and weak side).

ML Signals (5 models fully tested)

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	Beta vs SPY
Signal 2 (Ensemble) ★	**0.806**	**9.62%**	12.32%	**−4.59%**	0.275
Signal 4 (PCA+Ridge)	−0.083	−1.53%	11.20%	−7.98%	—
Signal 5 (MLP)	−0.388	−4.12%	9.67%	−10.40%	—
Signal 3 (XGBoost)	−0.468	−3.38%	6.85%	−10.76%	—
Signal 1 (LightGBM)	−1.681	−9.85%	6.05%	−13.37%	—

ML signals perform significantly worse in the L/S framework. Signal 2 is the only model with positive Sharpe (0.806); the remaining four all lose money. This indicates that current ML signals lack sufficient **short-side predictive accuracy** — identifying which sectors will underperform is harder than identifying outperformers.

VI. Overall Comparison & Key Conclusions

6.1 Full Strategy Comparison

Framework	Signal	Sharpe	Ann. Return	Max DD	Beta vs SPY	CAPM Alpha
L/S Baseline	**Alpha#23**	**2.081**	**23.00%**	**−5.85%**	**0.234**	**17.33%**
Softmax Long-Only	Alpha#19	1.968	23.33%	−7.26%	0.613	11.35%
Softmax Long-Only	Signal 2 (ML)	1.506	16.76%	−10.27%	0.691	4.52%
Equal-Weight	**—**	**1.305**	**13.98%**	**−10.64%**	**0.674**	**2.37%**
L/S Baseline	Signal 2 (ML)	0.806	9.62%	−4.59%	0.275	5.33%

6.2 Three Core Conclusions

① L/S + Alpha#23 is the best current configuration

Sharpe 2.08, max drawdown only −5.85%, market Beta 0.23. Returns come almost entirely from cross-sectional sector calls, not from passive market exposure. This is a textbook **pure-Alpha strategy** profile. CAPM Alpha of 17.33% vs SPY.

② Formulaic alphas significantly outperform ML signals in the L/S framework

Price-structure alphas capture structural price patterns and have stable discriminatory power on both long and short sides. Current ML signals lack adequate two-way predictive accuracy — further improvement is needed in Steps 2 and 3.

③ Signal quality is framework-specific — results cannot be cross-applied

The best Softmax signal (Alpha#19, Sharpe 1.968) flips to Sharpe −0.59 in the L/S framework. The best L/S signal (Alpha#23, Sharpe 2.08) scores only 1.665 in Softmax. **Signal screening must be conducted within the**

target execution framework — cross-framework conclusions are not transferable.

VII. Next Research Directions

7.1 Multi-Signal Combination Optimisation (Step 2)

Step 1 has demonstrated that Alpha#23 (L/S framework) and Signal 2 (long-only framework) are each effective individually. The next step is to **linearly blend multiple signals**:

$$s_{\text{combined}} = \alpha_1 \cdot g_{\text{Alpha#23}} + \alpha_2 \cdot g_{\text{Signal 2}} + \cdots$$

Bayesian optimisation searches for optimal weight combinations, paired with **walk-forward out-of-sample validation** to prevent overfitting.

7.2 Neural Network End-to-End Learning (Step 3)

Feed all 82 factors as input directly into a neural network to learn optimal output weights, with the entire backtest Sharpe as the training objective. This is the highest form of signal optimisation, but requires more rigorous overfitting controls.

> **Research Limitation**
>
> All backtest results in this report are produced under **zero transaction fees and zero slippage** assumptions. This is intentional: to purely evaluate signal predictive power, removing transaction cost interference. In live trading, costs (especially for high-turnover strategies) will materially reduce Sharpe. Current results do not represent achievable live returns. Live feasibility requires a dedicated cost-sensitivity study.

This report is based on backtest data from 2025-01-01 to 2026-01-31. Results are for research purposes only and do not constitute investment advice.